SECJ3553 ARTIFICIAL INTELLIGENCE

Section 03

ASSIGNMENT 1 (3%)

Theme: Agriculture

AI-Powered Predictive Analytics for Optimised Crop Yield and Risk Mitigation

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PREDICATES

1.1 Predicates Definition

Predicates	Description
Is the crop old enough to be harvested? **HarvestableAge()**	This predicate checks whether the crop is old enough to be harvested in an emergency. For example, 80-day-old rice field is not perfect but it is old enough to be saved from the flood.
Is the crop at its perfect age to be harvested? **OptimalAge()**	This predicate checks whether the crop is at the best age for the maximum and highest quality harvest. For rice fields, it is between 105 and 120 days.
Is there a high flood risk? **FloodRiskHigh()**	This predicate checks the weather forecast for the high risk of floods in the next 3 days. If it sees the risk, it will suggest an emergency harvest.
Is the weather safe for harvesting? **WeatherRiskLow()**	This predicate checks whether the weather is safe and there is no heavy rain or storm. This helps farmers to know that now is a good time to harvest.
Is the humidity good for planting? **HumidityOptimal()**	This predicate checks whether the humidity is suitable for planting in the next 2 days. Good humidity helps seeds grow well.
Is the farm plot empty? **PlotEmpty()**	This predicate checks whether a piece of land is vacant and new seeds are ready. This prevented the system from telling farmers to plant on top of other crops.
Suggest Early Harvest **SuggestEarlyHarvest()**	This is an emergency alert that tells farmers to harvest now to protect crops from bad weather, such as floods.
Suggest Best Harvest **SuggestOptimalHarvest()**	This is a signal to tell farmers that crops are at the best age and the weather is good, so they should harvest now for the best results.
Suggest Planting Now **SuggestPlantNow()**	This alarm tells the farmers that the plot is empty and the humidity is perfect, so now is the best time to sow.
Suggest to Wait **SuggestWait()**	This is a message to let farmers know that no action is needed now. It shows that the system is still monitoring, but everything is fine.
Check the Yield Model **CheckYieldModel()**	This tells the system to use its artificial intelligence model to guess how many crops farmers will get. It checks the age, type and weather of the crop, and then makes a prediction on the farmer's screen.

KNOWLEDGE REPRESENTATION

2.1 Rule 1: Emergency Harvest Suggestion

Production rules:

IF a crop is of a harvestable age, **AND** there is a high flood risk forecast within the next 3 days, **THEN** the agent must suggest an 'Early Harvest' advice to save the crop.

Explanation:

This is the most important part of the system, because it directly helps to prevent crop losses caused by floods. It checks two main conditions. It confirms that the crop has reached the harvestable age, which means that it is old enough to be preserved, just like rice that has been more than 80 days old. It also checks the danger by checking whether the weather API is predicting the high flood risk. If both conditions are true, the agent will recommend suggestions, which will display an alert on the farmer's dashboard.

Predicate logic/ First-order-logic (FOL):

$$\forall \mathrm{crop}[\mathrm{HarvestableAge}(\mathrm{crop}) \wedge \mathrm{FloodRiskHigh}() \rightarrow \mathrm{SuggestEarlyCrop}()]$$

2.2 Rule 2: Optimal Harvest Suggestion

Production rules:

IF a crop's age is in the ‘Optimal Harvest’ for its type, **AND** the weather forecast shows 'Low' risk, **THEN** the agent will suggest an ‘Optimal Harvest’.

Explanation:

This rule tells the farmer the best time to harvest to get most yield and quality. For example, it will predict the yield and tell the farmer like ‘Paddy is best harvested between 105 and 120 days. The rule also checks that the weather forecast show ‘Low’ risk, so the farmer is not told to harvest during a storm. So, if the crop is the right age and weather is safe, it recommends an ‘Optimal Harvest’.

Predicate logic/ First-order-logic (FOL):

$$\forall \mathrm{crop}[\mathrm{OptimalAge}(\mathrm{crop}) \wedge \mathrm{WeatherRiskLow}() \rightarrow \mathrm{SuggestOptimalHarvest}()]$$

2.3 Rule 3: Predicted Yield Calculation

Production rules:

IF the agent is recommending any harvest action, either ‘Early Harvest’ or ‘Optimal Harvest’, **THEN** it must also check the ‘Yield Model’ to predict the expected yield for that crop at its current age.

Explanation:

This rule makes money for the farmer. It links the suggestions provided by the agent to a money result. When the system makes a harvest suggestion using Rule 1 or Rule 2, this rule runs at the same time. It takes the crop's current age, type, and data like weather and feeds it into a trained machine learning model. The model is trained on a historical dataset, then predicts the most likely yield. For the logic, this entire model is represented as a single predicate. The resulting prediction is then shown on the dashboard.

Predicate logic/ First-order-logic:

$$\forall \mathrm{crop}[\mathrm{SuggestEarlyHarvest}(\mathrm{crop}) \vee \mathrm{SuggestOptimalHarvest}(\mathrm{crop}) \rightarrow \mathrm{CheckYieldModel}(\mathrm{crop})]$$

2.4 Rule 4: Optimal Planting Recommendation

Production rules:

IF a farmer's plot is ‘Empty’, **AND** the humidity forecast for the next 2 days is in the ‘Optimal Planting’ for the crop they want, **THEN** the agent will suggest ‘Plant Now’.

Explanation:

This rule helps the farmer start their crop in the best time. It checks that the land is ready using information about the plot's current status. It will see the best humidity range for planting the chosen crop type and compares this to the humidity forecast. If the ground is empty and predicted humidity is right, then the agent will recommend ‘Plant Now’ to make sure the crop is planted when it has the best chance to grow well.

Predicate logic/ First-order-logic:

$$\forall \mathrm{plot}, \mathrm{crop}[\mathrm{PlotEmpty}(\mathrm{crop}) \wedge \mathrm{HumidityOptimal}(\mathrm{crop}) \rightarrow \mathrm{SuggestPlanNow}(\mathrm{crop})]$$

2.5 Rule 5: Wait or No Action

Production rules:

IF a crop is not in the best harvest time, **AND** it is not old enough for an emergency harvest, **OR** the weather is not ‘Low’ risk for harvesting, **THEN** the agent will recommend to ‘Wait’.

Explanation:

This is the safe rule. It is the opposite of the other harvest rule. It makes sure that if no action like ‘Harvest Early’ or ‘Optimal Harvest’ is needed, the agent will tell the farmer to ‘Wait’. This rule will stop confusion and let the farmer know the system is investigating the condition, but nothing is needed from the farmer’s side right now.

Predicate logic/ First-order-logic:

$$\forall \mathrm{crop}((\neg \mathrm{OptimalAge}(\mathrm{crop}) \wedge \neg \mathrm{HarvestableAge}(\mathrm{crop})) \vee \neg \mathrm{WeatherRiskLow}()) \rightarrow \mathrm{SuggestWait}(\mathrm{crop})]$$

KNOWLEDGE REPRESENTATION JUSTIFICATION

We use Production Rules and Predicate Logic (First-Order Logic, FOL) to hold the knowledge in this smart agriculture forecasting system. We chose them because they give us a clear and easy to understand way to make decisions. This method lets the system hold expert knowledge in a form that an AI agent can easily use for reasoning.

3.1 Use of Production Rules

We choose production rules because they are easy to read. This table is like a farm expert making a decision according to specific conditions. For example, if the crop has reached the harvestable age and the risk of flooding is high, it is recommended to harvest as early as possible. This method has several advantages. It's easy to update, so people can add new farm rules without having to redesign the entire system. It also provides transparent reasoning, which means that farmers can understand why the system advises them. Finally, it is suitable for real-time decision-making, because the system can check many situations at the same time, such as crop age or weather risk.

3.2 Use of Predicate Logic / First-Order Logic

We use predicate logic to write rules in a mathematically understandable way that machines can understand. It provides us with a very precise way to show the relationship between crops, farm plots, weather forecasts and actions. By using words ($\forall$, $\exists$) and logical operators ($\land$, $\lor$, $\neg$, $\rightarrow$), the system can make general statements. For example, for all crops, if the crops can be harvested and the risk of flooding is high, it is recommended to harvest as early as possible. This makes sure all the rules work together. It lets the system find new answers when it gets new data. It also shows how different things are connected, such as how a crop is affected by weather rules. Most importantly, it makes the rules very clear and simple, so there will be no confusion.

3.3 Why These Representations Fit the Domain

Making a farm decision means considering the weather, the age of the crops, and the degree of soil moisture. Only after checking everything can farmers decide whether to harvest or wait. We use production rules to carry out this if-then thinking which is very suitable. Then, we use predicate logic to provide the system with a computer-friendly way to check all these rules. We use both methods to make the system better. It becomes modular, which means that each rule is a neat piece of its own, so the system remains clean and organised. This can also be explained, which means that the system can show you the exact rules it uses to give you advice. Finally, the system is expandible, which means that new crops, new weather rules or new information can be added to the system at any time.